MIL-C-3120F(EA)
14 February 1978
SUPERSEDING
MIL-C-3120E(MU)
12 July 1968

MILITARY SPECIFICATION

CANISTERS, SMOKE, HC, 155MM

PROJECTILE, M1 and M2

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the filling and assembly of HC Smoke Canisters for a base ejection shell.

1.2 Classification. The canister components shall be furnished in the following types as specified (see 6.2):

Type M1 - Canister, Smoke, HC, 155MM Projectile (Dwg D15-11-33).

Type M2 - Canister, Smoke, HC, 155MM Projectile (Dwg D15-11-41).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 1320

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistance (For Carton Sealing).

MILITARY

MIL-B-117 - Bags, Sleeves, and Tubing, Interior Packing.
MIL-P-223 - Powder, Black.
MIL-A-2550 - Ammunition, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads.
MIL-STD-650 - Explosive; Sampling, Inspection and Testing.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1234 - Pyrotechnic, Sampling, Inspection and Testing.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

*CHEMICAL SYSTEMS LABORATORY

TDPL-15-11-33 - Canister, Smoke, HC, 155MM Projectile, M1.
TDPL-15-11-41 - Canister, Smoke, HC, 155MM Projectile, M2.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*Formerly Edgewood Arsenal.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL-15-11-33 and TDPL-15-11-41 shall conform to the specifications listed thereon, unless modified specific characteristics are set forth on the drawings, in which case, detail drawings take precedent (see 6.4).

3.1.2 Components. All components of the M1 and M2 canisters shall conform to the specifications and drawings listed on TDPL-15-11-33 and TDPL-15-11-41.

3.2 Filling and assembly. The M1 and M2 canisters shall be filled and assembled as specified on Drawings D15-11-118, D15-11-33 (M1) and D15-11-123, D15-11-41 (M2) respectively (see 6.4 and 6.5). The white smoke mixture (HC), (Drawing B143-1-1) shall be consolidated into the body assembly, after the impregnated ignition sleeve, starter slugs/zinc sleeve assembly, and packing disk are in place, in two increments for the M1 canister and one increment for the M2 canister. Each increment shall be consolidated at 80 - 82 tons dead load for a period of not less than 3 seconds for the M1 canister. The M2 canister increment shall be consolidated at 70 - 72 tons dead load for a period of not less than 3 seconds dwell time. The excess white smoke mixture shall be removed to a height as shown on Drawings D15-11-118 (M1) and D15-11-123 (M2) respectively. Additional packing disks, as required, and the closing disk shall be installed and the body assembly completed as indicated on the Drawings D15-11-118, D15-11-33 (M1) and D15-11-123, D15-11-41 (M2) respectively.

3.3 Starter slug.

3.3.1 Volatile content. The volatile content of the finished starter slug shall not exceed 0.50 percent when tested in accordance with 4.4.4.2(a).

3.3.2 Crush strength. The crush strength of the starter slug shall be such that the starter slug shall not crush or break when a uniformly distributed load of 500 lbs is applied parallel to the longitudinal axis of the starter slug and when tested in accordance with 4.4.4.2(b).

3.3.3 Starter mixture. The starter mixture shall conform to the chemical composition specified on Drawing C15-11-119 when tested in accordance with 4.4.4.2(c).

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3.4 Impregnated sleeve.

3.4.1 Burn time. The impregnated sleeve shall conform to Drawings B15-11-125 and B15-11-127. The dried impregnated sleeve shall demonstrate complete burning over its entire area within 7 seconds maximum when tested in accordance with 4.4.4.3(a).

3.4.2 Impregnating mixture. The impregnating mixture shall conform to the chemical composition specified on Drawing B143-15-1 when tested in accordance with 4.4.4.3(b).

3.5 HC mixture. The HC mixture shall conform to the chemical composition specified on Drawing B143-1-1 when tested in accordance with 4.4.4.4.

3.6 Functioning. The M1 canister shall produce a continuous dense cloud of white smoke (opaque) for 60 to 150 seconds for the M1 and 30 to 120 seconds for the M2, when tested as specified in 4.4.4.1. Time from the first ignition of starter mix to first emission of smoke under pressure shall not exceed 10 seconds. Smoke emission time shall be that period, when smoke is emitted under pressure until the cessation of smoke under pressure. Smoldering of the canister after the charge has burned is not considered part of the smoke emission time. There shall be no interruption of smoke by flame (burning of the starter mixture, first 10 seconds maximum, shall not be construed as flaming).

3.7 First article ballistic information. This specification makes provisions for a first article ballistic informational test prior to commencement of regular production. For test requirements see 4.3.4 and Table I.

3.8 Preproduction. Prior to the start of regular production, a preproduction lot of canisters shall be produced in accordance with this specification, for examination and tests (see 4.3).

3.9 Workmanship. The canister shall be free from contamination (pyrotechnic mixture, oil, or grease) and damage, including but not limited to dents, cracks, or deformation which could affect its intended use. The starter slug shall be free from contamination (foreign matter, oil, or grease) and damage including but not limited to cracks, chip outs, dents, or deformation which could affect its assembly into the canister or its intended use. The impregnated sleeve shall be free from contamination (foreign matter, oil, or grease) and damage including, but not limited to, tears, rips, frayed areas, and deformation which could affect its assembly into the canister or its intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the inspection where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The provisions of MIL-A-2550 shall apply.

4.1.2 Additional responsibility of contractor.

(a) Drawings for contractor designed inspection equipment (other than standard measuring equipment) shall be forwarded to design agency for approval prior to production (see 6.2).

(b) Ammunition data cards prepared in accordance with MIL-STD-1167 shall be supplied for each lot of canisters.

(c) The contractor's quality assurance element will establish controls sufficient to assure compliance with requirements of starter mix Dwg C15-11-119, impregnating mix Dwg. B143-15-1, HC mix Dwg B143-1-1, and Section three of this specification. The QA plan shall be approved by the procuring activity prior to initiation of production.

(d) The sampling of mix batches/lots for the weighing and blending operations shall be in accordance with the requirements of MIL-STD-1235, CSR1, AQL 6.5. The quality assurance representative shall witness the weighing and blending of every sample batch or lot.

(e) A batch/lot record specifying the weight of each chemical ingredient shall be maintained for each blended batch/lot produced.

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 1,250 canisters, 850 M1 and 400 M2, shall be manufactured using the same methods, materials, equipment, processes, and cycle time as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests.

(a) Examination. Sampling of canisters of each type for examination shall be conducted in accordance with MIL-STD-105, level II.

(b) Filling and assembly. Sampling of canisters of each type for examination in accordance with 3.2, (Drawings D15-11-118 and D15-11-123) shall be conducted in accordance with MIL-STD-105, level II and classification of defects (4.4.3.3).

(c) Volatile content of starter slugs. A minimum of two starter slug samples shall be selected from each batch (see 6.5) for test in accordance with 4.4.4.2(a).

(d) Impregnated sleeve. A random sample of 5 each sleeves shall be taken from each batch/lot of sleeves.

4.3.2.2 For destructive tests.

(a) Functioning. Sampling of canisters of each type shall be conducted in accordance with MIL-STD-105, level S-4.

(b) Crush strength of starter slug. Three starter slugs per batch/lot shall be taken at random for the crush strength test in accordance with 4.4.4.2(b).

(c) Impregnated sleeve. A random sample of 5 each sleeves shall be tested in accordance with 4.4.4.3(a).

(d) Starter mix. Select at random one sample from every eight batches/lots of starter mixture used to manufacture slugs.

(e) HC mix. Select at random one sample from every eight batches/lots of HC smoke mix produced.

(f) Impregnating mix. Select at random one sample from every eight batches of impregnating mix produced.

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(g) First article ballistic sample. A random sample of 135 M1 and 45 M2 canisters shall be taken from the preproduction lot and submitted to a Government Proving Ground as designated by the contracting officer (see 6.2).

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests.

(a) Examination. Sample canisters of each type shall be examined in accordance with 4.4.3.3 and MIL-STD-105.

(b) Filling and assembly. Sample canisters of each type for filling and assembly shall be examined in accordance with 3.2 (Drawings D15-11-118 and D15-11-123) and classification of defects (4.4.3.3).

(c) Volatile content of starter slug. The starter slugs shall be tested in accordance with 4.4.4.2(a). The acceptance number is zero.

(d) Impregnated sleeve. Five sleeves of each type shall be examined in accordance with Drawings B15-11-125 and B15-11-127. The acceptance number is zero.

4.3.3.2 For destructive tests.

(a) Functioning. Sample canisters of each type shall be tested in accordance with 4.4.4.1. The AQL shall be as specified in 4.4.3.2(a).

(b) Crush strength of starter slug. Three starter slugs per batch/lot shall be tested for crush strength in accordance with 4.4.4.2(b). The acceptance number is zero.

(c) Impregnated sleeve. Five sleeves of each type shall be tested in accordance with 4.4.4.3(a). The acceptance number is zero.

(d) Starter mix. The sample starter mix shall be tested in accordance with 4.4.4.2(c). If any sample fails the requirement, the lot represented shall be rejected. In addition, all lots subsequent to the last acceptable lot sampled, shall be analyzed for conformance to Dwg C15-11-119. Lots not meeting dwg requirements shall be rejected. Slugs produced from batches/lots of starter mix found to be defective shall be rejected.

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(e) HC mixture. The sample of HC smoke mix shall be qualitatively analyzed for conformance to Dwg B143-1-1. If the sample fails to meet drawing requirements, the lot shall be rejected. In addition, all lots subsequent to the last acceptable lot samples shall be analyzed for conformance to Dwg B143-1-1, all lots not meeting dwg requirements shall be rejected.

(f) Impregnating mixture. The sample of impregnating mixture shall be qualitatively analyzed for conformance to Dwg B143-15-1. If the sample fails to meet Dwg requirements, the lot/batch shall be rejected. In addition, all lot/batches subsequent to the last acceptable sample shall be analyzed for conformance to Dwg requirements. All lots/batches not meeting Dwg requirements shall be rejected.

(g) First article ballistic functioning. Each type of canister shall be assembled and tested in accordance with 4.3.4. The contractor shall obtain written approval form the contracting officer prior to proceeding with regular production.

4.3.4 First article ballistic test. The sample canisters (4.3.2.2(g)) received by the proving ground shall be assembled into 45, 155MM, M116E2 HC smoke projectiles in accordance with drawing 8885162. Fifteen projectiles shall be temperature conditioned at +70° ± 2.5° F, fifteen at -45° +5° F and fifteen at +145° -5° for 12 hours minimum at each temperature. After temperature conditioning, each projectile shall be fired within 5 minutes of removal from the temperature conditioning in a howitzer equipped with an M126 or M1A2 cannon utilizing the standard M4A2 propelling charge conditioned at +70° + 2.5°F. Each round shall be fired at an elevation and fuze setting that will permit ejection of all four canisters in flight and subsequent observation of smoking canisters. The desired canister fall time is 5 seconds, minimum. Appropriate firing tables should be consulted and sufficient stock rounds fired for spotting to verify correctness of elevation and fuze setting. Observation shall be made for the performance of the shell in accordance with table I. In addition, muzzle velocity, chamber pressure, range (canister impact), and time of flight for each round shall be recorded. A smear camera shall be used to observe the projectile in flight at approximately 25 feet in front of the weapon to record any possible malfunction. Copies of the official firing record shall be sent to the Government Agency designated by the contracting officer (see 6.2).

Table I. Malfunctions

a. Misfires. The projectile fails to fire from the weapon.

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b. Premature functioning and metal parts failure. There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 200 yards (182.88 meters) of horizontal range.

c. Fuze dud. Absence of visible evidence of fuze functioning as evidenced by a flash or presence of a smoke cloud.

d. Projectile dud. Canister payload failed to eject.

e. Lost. Observer(s) fail to see canister ejection or canister burning.

f. Canister dud. The canister is ejected but no smoking or burning occurs.

g. Exploder. The canister is ejected and then explodes. This may occur with or without some normal smoke emission. It generally results from smoke mix anomaly and causes a black cloud at the point of incident. The metal parts commonly have the canister lid and/or bottom blown out.

h. Disintegrator. The canister is ejected and then splits open (usually into approximately equal segments), dispersing the smoke mix into the air. The cloud thus produced is the same color as the fill. The metal parts usually have the canister lid on the flash tube.

i. Delayed ignition. The smoke emission from the canister did not begin within 12 seconds.

j. Incorrect burn time. The canister smoke emission time is less than 57 or more than 170 seconds for the M1 canister and less than 27 or more than 140 seconds for the M2 canister.

k. Poor smoke. HC smoke that does not obscure.

l. Flamer. Canister flaming greater than 10 percent of smoke emission time. Canisters that fall in tall, dry grass, will be observed very carefully to insure that the grass does not burn first and then flame the canister. HC canisters exhibit flame at each end during complete smoke emission time, but this is not considered to constitute flaming.

4.3.5 Acceptance/rejection criteria. The preproduction samples shall comply with the examination and tests specified in 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168, and shall consist of canisters produced in no more than five consecutive work shifts not to exceed 2 weeks consecutive duration.

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Only one interfix number of components from one manufacturer and only one manufacturer's component shall be in one canister interfix number.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests.

(a) Examination. Sampling of canisters of each type for examination shall be conducted in accordance with MIL-STD-105, Level II.

(b) Filling and assembly. Sampling of canisters of each type for examination in accordance with the requirements of 3.2 (Drawings E15-11-118 and D15-11-123) shall be conducted in accordance with MIL-STD-1235, CSP-1, AQL 1.0%, except that sampling for the length, diameter, and crimp requirements for filled canisters may be in accordance with MIL-STD-105, Level II.

(c) Volatile content of starter slug. Two starter slugs, shall be selected at random from each batch/lot for test in accordance with 4.4.4.2(a).

(d) Impregnated sleeve. A random sample of 5 each sleeves shall be taken from each batch/lot of sleeves.

4.4.2.2 For destructive test.

(a) Functioning. Sampling of canisters of each type shall be conducted in accordance with MIL-STD-105, Level S-4.

(b) Crush strength of starter slug. Three starter slugs per batch/lot shall be used for the crush strength test in accordance with 4.4.4.2(b).

(c) Impregnated sleeve. A random sample of 5 each type sleeves shall be tested in accordance with 4.4.4.3(a).

(d) Starter mixture. Select at random one sample from every fifteen batches/lots of starter mixture used to manufacture slugs.

(e) HC mixture. Select at random one sample from every fifteen batches/lots of HC smoke mix produced.

(f) Impregnating mixture. Select at random one sample from every fifteen batches/lots impregnating mix produced.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests.

(a) Examination. Sample canisters of each type shall be examined in accordance with the classification of defects and MIL-STD-105, Level II.

(b) Filling and assembly. Sample canisters of each type for filling and assembly shall be inspected for requirements of 3.2 (Drawings D15-11-118 and D15-11-123) using MIL-STD-1235, CSP-1 using an AQL of 1.0 percent defective. The sample of filled canisters may be examined for length, diameter, and crimp in accordance with MIL-STD-105 and classification of defects (4.4.3.3).

(c) Volatile content of starter slug. The volatile content of starter slugs shall be determined in accordance with 4.4.4.2(a). The acceptance number is zero.

(d) Impregnated sleeve. Five sleeves of each type shall be examined in accordance with Drawings B15-11-125 and B15-11-127. The acceptance number is zero

4.4.3.2 For destructive tests. Sample canisters of each type shall be tested in accordance with 4.4.4.1 and MIL-STD-105. The AQL shall be as follows:

(a) Functioning. The smoke emission time, delay, and a dud* shall have an AQL of 1.0 percent defective.

*A dud is a canister which for any reason fails to function. No canister shall be counted as a defective more than once, regardless of how many defects that one canister exhibits.

(b) Crush strength of starter slug. Three starter slugs per batch/lot shall be used for the crush strength test (4.4.4.2(b)). The acceptance number is zero.

(c) Impregnated sleeve. Five sleeves of each type shall be tested in accordance with 4.4.4.3(a). The acceptance number is zero.

(d) Starter mix. The sample starter mix shall be tested in accordance with 4.4.4.2(c). If any sample fails the requirement, the lot represented shall be rejected. In addition, all lots subsequent to the last acceptable lot sampled shall be analyzed for conformance to Dwg. C15-11-119. Lots/batches not meeting requirements shall be rejected. Slugs produced from batches/lots of starter mix found to be defective shall be rejected.

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(e) HC mixture. The sample of HC smoke mix shall be qualitatively analyzed for conformance to Dwg. B143-1-1. If the sample fails to meet drawing requirements, the lot shall be rejected. In addition, all lots/batches subsequent to the last acceptable sample shall be analyzed for conformance to drawing requirements. All lots/batches not meeting drawing requirements shall be rejected.

(f) Impregnating mixture. The sample of impregnating mixture shall be qualitatively analyzed for conformance to Dwg. B143-15-1. If the sample fails to meet drawing requirements the lot/batch shall be rejected. In addition, all lots/batches subsequent to the last acceptable sample shall be analyzed for conformance to drawing requirements. All lots/batches not meeting drawing requirements shall be rejected. Sleeves produced from batches/lots of impregnating mix found to be defective shall be rejected.

4.4.3.3 Classification of defects. See pages 13 thru 17.

4.4.4 Tests. (Precaution Note. See 6.4)

4.4.4.1 Functioning (See 6.4) The test shall be conducted as follows:

Remove the squares of tape from the ends and insert a piece of primed time fuze (Bickford type) at least 6 inches long approximately 2 inches in the canister. For priming method, see note. Lay canister on its side (flash tube in horizontal position), and ignite the fuze with a fuze lighter or similar device. The canister shall be functioned in an area clear of flammable material, grass, etc., that may influence flaming. If fuze fails to ignite the canister, refuzing will be alloved one time only. If canister fails to light with the second fuze, the canister shall be considered a dud.

NOTE: Preparation of primed fuze.

A 6-8 inch length of safety fuze (Class 1375, Bickford type) 0.200 ± .005 inch O.D., burning time 120 seconds per yard ± 10 percent, shall be primed on one end between 1 and 1-1/2 inches by dipping in a slurry composed of starter mixture with binder (Drawing C15-11-119) or a slurry composed of black powder (Class 5, MIL-P-223) and unplasticized collodion (See 6.6). The fuze may be dipped more than once, depending on the thickness of the slurry, to assure adequate priming. The fuze shall be air dried for a minimum of 2 hours prior to use.

4.4.4.2 Starter slugs.

(a) Volatile content. The volatile content shall be determined for each batch of starter slugs in accordance with MIL-STD-650, Method 101.6 Moisture and Volatiles (Vacuum Oven Method).

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 1 of 5		DRAWING NUMBER
4.4.3.3	Canister, Smoke, HC, 155MM Projectile, M1 and M2					B15-11-126 and B15-11-128 Starter Slug
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH		NEXT HIGHER ASSEMBLY D15-11-118, D15-11-123 PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined					
Major:						
101	Inside diameter incorrect		0.40%	3.1.2		Gage or Standard Measuring Instruments (SMI)
102	Outside diameter incorrect		0.40%	3.1.2		Gage or SMI
103	Length incorrect		0.40%	3.1.2		Gage or SMI
104	Workmanship		0.40%	3.9		Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 of 5		DRAWING NUMBER
4.4.3.3	Canisters, Smoke, HC, 155MM Projectile, M1 and M2				Body Assembly, Filled D15-11-118(M1) and D15-11-123(M2)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY D15-11-33, D15-11-41 PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Component missing		0.40%	3.2	Visual
102	Diameter incorrect		0.40%	3.2	Gage
103	Length incorrect		0.40%	3.2	Gage
104	Crimp incorrect		0.40%	3.2	Feeler gage
105	Filling faced height incorrect (check before insertion of closure disk).		0.40%	3.2	Gage
106	Cavities excessive (chipouts, broken edges etc. - check before insertion of closure disk)		0.40%	3.2	Visual
107	Workmanship		0.40%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 of 5		DRAWING NUMBER
4.4.3.3	*Canisters, Smoke, HC, 155MM Projectile, M1 and M2				D15-11-33(M1) and D15-11-41(M2)*
					NEXT HIGHER ASSEMBLY 8885162
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Marking missing, illegible or incorrect		0.40%	3.2	Visual
102	Exterior finish incorrect or missing		0.40%	3.2	Visual
103	Tape missing		0.40%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE				DRAWING NUMBER
4.4.3.3	Canisters, Smoke, HC, 155MM Projectile M1 and M2		SHEET 4 of 5		B15-11-125 and B15-11-127(M2) Sleeve, Ignition
					NEXT HIGHER ASSEMBLY D15-11-118(M1), D15-11-123(M2)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Length incorrect		0.40%	3.1.2	SMI
102	Overlap incorrect		0.40%	3.1.2	Visual
*103	Inside diameter incorrect		0.40%	3.1.2	Tool control
104	Workmanship		0.65%	3.9	Visual
	*Tool control: Tooling to be checked prior to and upon completion of production or any time tooling is modified or changed. Tooling must also be verified after each 40 hours of operation. Tooling must be correct at each check.				

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER Preparation for delivery Section 5 of MIL-C-3120
4.4.3.3	Canisters, Smoke, HC, 155MM Projectile M1 and M2				NEXT HIGHER ASSEMBLY TDPL 15-11-33(M1) 15-11-41(M2)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Packing wood not preserved or not dry (Level A Pack only)		0.40%	5	Test per PPP-B-621
102	Canister packed incorrectly		0.40%	5	Visual
103	Loose pack		0.40%	5	Visual
104	Component missing		0.40%	5	Visual
105	Strapping missing, loose, or incorrect		0.40%	5	Visual
106	Marking illegible or incorrect		0.40%	3 and 5	Visual
107	Box damaged or improperly assembled		0.40%	5	Visual
108	Barrier bag improperly closed or damaged		0.40%	5	Visual

NOTES:

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(b) Crush strength. The crush strength test load on the starter slug shall be applied uniformly parallel to the longitudinal axis at a constant rate of 1/4 inch per second maximum.

(c) Starter mixture. The sample of starter mixture shall be qualitatively analyzed for conformance to Drawing C15-11-119.

4.4.4.3 Impregnated sleeve.

(a) Impregnated sleeve. The impregnated sleeve shall be laid on its side clear of flammable material and ignited at one end with a minimum heat source necessary for ignition. Once ignited, the sleeve shall burn as specified.

(b) Impregnating mixture. The impregnating mixture shall be qualitatively analyzed for conformance to Dwg. B143-15-1.

4.4.4.4 HC mixture.

(a) The moisture content shall be determined for each batch/lot of HC Smoke mix in accordance with MIL-STD-1234 method 101.2.1.

(b) The samples of HC mix shall be qualitatively analyzed for conformance to Dwg. B143-1-1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A . Twelve canisters of the same type shall be packaged in a fiberboard box conforming to grade W5c, style RSC of PPP-B-636. Inside nominal dimensions of the box shall be as follows:

(a) M1 Canister 19 by 14-1/4 by 5 inches, length (l) by width (w) by depth (d) (l by w by d).

(b) M2 Canister 17-1/2 by 13-1/8 by 4 inches, (l by w by d).

5.1.1 Partition cells. Arrangement of the canisters shall be four rows of three canisters in each. Interlocking partitions shall separate the canisters from one another and shall be constructed of material conforming to grade W5c of PPP-F-320. The partition cell nominal measurements shall be as follows:

(a) M1 Canister 4-5/8 by 4-5/8 by 4-7/8 inches (l by w by d).

(b) M2 Canister 4-1/4 by 4-1/4 by 3-7/8 inches (l by w by d).

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5.1.2 Sealing. The fiberboard box shall be closed in accordance with requirements of PPP-B-636, using PPP-T-76 tape. The package shall be sealed within a barrier bag conforming to type II, class E, style 1 of MIL-B-117, utilizing the spun bonded polyolefin backing material, and having nominal measurements as follows:

(a) M1 Canister 26 by 22 inches (l by w) (useable area).

(b) M2 Canister 24 by 20 inches (l by w) (useable area).

Included air volume shall be evacuated to a vacuum of 25 inches of water until the bag conforms to the fiberboard box. Wrinkling of barrier materials shall be held to a minimum.

5.2 Packing.

5.2.1 Level A . Canisters packaged as specified in 5.1, shall be packed in a wooden shipping container conforming to style 4, class 2, grade A, type 2 load of PPP-B-621. Weight limitations for construction of the shipping container shall be within the 100 to 250 pound category. Fiberboard liners conforming to grade W5c of PPP-F-320, shall be placed within the shipping container to separate the contents from the top, bottom, sides, and ends of the shipping container. Additional fiberboard filler pads shall be used, as required, to provide a tight pack. The normal inside dimensions of the shipping container shall be as follows:

(a) M1 Canister - 19-3/4 by 15 by 5-1/2 inches (l by w by d).

(b) M2 Canister - 18-1/4 by 13-7/8 by 4-1/2 inches (l by w by d).

After loading, the shipping container shall be closed and strapped in accordance with the container specification, using flat, galvanized steel strapping conforming to QQ-S-781.

5.2.2 Interplant shipment. For Continental U.S. (Outside Continental US comply with paragraph 5.1 and 5.2) - (See 6.3). Twelve canisters of the same type shall be packed together in one layer in a wooden shipping container conforming to PPP-B-621, style 3, class 1, grade B, type 2 load for domestic shipment, weight limitations for construction of the box shall be in the 100-250 pound category. Case liners and interlocking partitions are not required. Normal inside dimensions of the wood box shall be as follows:

(a) M1 Canister - 18-3/4 by 14-1/8 by 5-1/8 inches (l by w by d).

(b) M2 Canister - 17-1/4 by 13 by 4-1/8 inches (l by w by d).

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Fiberboard filler pads, conforming to grade W5c of PPP-F-320, shall be used, as required, to assure a tight pack. After loading, the wooden shipping container shall be closed by strapping with two each, 3/8 inch, class C, flat steel strapping conforming to QQ-S-781. Nailing of the lids is not required.

5.3 Marking. In addition to any special marking as may be required by the contract or order, wooden boxes shall be marked in accordance with the applicable requirements of MIL-STD-129. The shipping container lid shall be marked with 3/4 inch (minimum) letters as follows:

SMOKE SIGNALS, HANDLE CAREFULLY, KEEP FIRE AWAY

5.4 Palletization. When specified in the contract or order, boxes packed as specified in 5.2.1 shall be palletized in accordance with the applicable method of MIL-STD-147.

6. NOTES

6.1 Intended use. The HC smoke canisters are components for a 155MM base ejection smoke shell, which is used for screening, spotting and signaling purposes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Design agency where drawings for contractor designed inspection equipment is to be forwarded.

(c) Government agency where first article sample is to be forwarded.

(d) Government agency where official firing record is to be forwarded.

(e) Canister type, (M1 or M2), quantity ordered.

(f) Preproduction:

(1) Time allowed for contractor's submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

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(g) Special marking required.

(h) When palletization is required.

6.3 Interplant shipment. Packaging for materials that do not enter the military supply system. Typical are shipments from a vendor to a contractor, a subcontractor, or a supplier to a military installation or plant.

6.4 Precaution. This specification covers sampling and testing of chemical toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials. Standard safety precaution procedures should be taken to prevent fires during preparation of the mixtures and while assembling canisters. Appropriate safety apparel should be worn when handling these materials. Caution should also be exercised during the functioning test as a hazard exists from flying hot particles and possible detonation resulting from incorrect mixture. Manufacture of this item involves pyrotechnic hazards. Potential contractors are advised to be knowledgeable in the art of handling and processing these hazardous items. Also see Major Component Safety Data Sheets, SDS No's. 20026, 20019, 20020, and 20028.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation to make the final product substantially uniform. A starter mixture batch shall not exceed 30 lbs.

6.6 Collodion. Unplasticized collodion may be procured from a laboratory supply company.

Custodian:

Army - (EA)

Preparing activity:

Army - (EA)

Project No. 1320-A403

☆U.S. GOVERNMENT PRINTING OFFICE: 1978 — 703-122/1472

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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